

Don't try to memorize this file line-by-line. Instead, remember the **Data Analysis Workflow**. Nearly every Pandas project follows this pattern:

1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Why?

- pandas → tables (DataFrames)
- numpy → calculations
- matplotlib → plotting
- seaborn → prettier plots

Remember: Import → Load → Explore → Clean → Analyze → Visualize

2. Load Data

```
df = pd.read_csv('netflix_titles.csv')
```

What happens?

Creates a DataFrame called `df`.

Think:

```
CSV File
  ↓
pd.read_csv()
  ↓
DataFrame (df)
```

3. Explore the Data

First 5 rows

```
df.head()
```

Shows:

Row 1
Row 2
Row 3
Row 4
Row 5

Purpose:

- Understand columns
- Check format

Shape

`df.shape`

Output:

`(8807, 12)`

Means:

8807 rows
12 columns

Remember:

`rows, columns`

Missing Values

`df.isnull().sum()`

Think:

`isnull()`
↓
True/False

`sum()`
↓
Count Trues

Example:

Director

John

NaN

Mike

Result:

```
1 missing value
```

Data Types

```
df.dtypes
```

Shows:

```
object  
int64  
float64  
datetime64
```

Meaning:

Type	Meaning
object	text
int64	integer
float64	decimal
datetime64	date

Duplicates

```
df.duplicated().sum()
```

Checks repeated rows.

Example:

```
Movie A  
Movie B  
Movie A
```

Result:

```
1 duplicate
```

Unique Values

```
df['country'].unique()
```

Output:

```
USA  
US  
United States  
India  
UK
```

Purpose:

Find inconsistent names.

4. Data Cleaning

Think:

```
Missing Values  
↓  
Wrong Names  
↓  
Wrong Data Types  
↓  
Ready Data
```

Fill Missing Director

```
df['director'] = df['director'].fillna('Unknown')
```

Before:

Director

NaN

After:

Director

Unknown

Remove Missing Ratings

```
df.dropna(subset=['rating'], inplace=True)
```

Means:

"Delete rows where rating is missing."

Standardize Country Names

```
df['country'] = df['country'].replace(  
    ['USA', 'US'],  
    'United States'  
)
```

Before:

```
USA  
US  
United States
```

After:

```
United States  
United States  
United States
```

Convert Date

```
df['date_added'] = pd.to_datetime(  
    df['date_added'],  
    errors='coerce'  
)
```

Before:

```
"September 9, 2019"
```

After:

```
2019-09-09
```

Now Python understands it's a date.

Extract Number From Duration

Before:

```
90 min
2 Seasons
5 Seasons
```

Code:

```
df['duration_int'] = (
    df['duration']
    .str.extract(r'(\d+) ')
    .fillna(0)
    .astype(float)
)
```

Output:

```
90
2
5
```

Breakdown

Extract digits

```
.str.extract(r'(\d+) ')
```

Regex:

```
\d = digit
+ = one or more
```

Extracts:

```
90 min → 90
2 Seasons → 2
```

5. Feature Engineering

This means:

Create new columns from existing columns.

Count Actors

```
df['num_of_actors'] =  
df['cast'].str.split(',').str.len()
```

Example:

Tom, Brad, John

Split:

```
['Tom', 'Brad', 'John']
```

Length:

3

Extract Year

```
df['year_added'] =  
df['date_added'].dt.year
```

Example:

2020-08-15

Output:

2020

Extract Month

```
df['month_added'] =  
df['date_added'].dt.month
```

Output:

8

Create Decade

```
df['decade'] =  
(df['release_year']//10)*10
```

Example:

2017

Step:

```
2017//10  
= 201
```

```
201*10  
= 2010
```

Result:

2010s

USA Flag

```
df['is_usa'] = np.where(  
    df['country']=='United States',  
    1,  
    0  
)
```

Output:

```
1 = USA  
0 = Not USA
```

First Genre

```
df['genre_simple'] =  
df['listed_in'].str.split(',').str[0]
```

Example:

Drama, Romance, Comedy

Output:

Drama

Check Keyword

```
df['has_netflix_in_title']
```

Created by:

```
str.contains('Netflix')
```

Example:

```
Netflix Originals
```

Result:

```
1
```

6. Summary Statistics

```
describe()
```

Shows:

```
count  
mean  
std  
min  
25%  
50%  
75%  
max
```

Example:

```
Average movie duration  
Average actors  
Average year
```

7. Visualization

This is where most students get confused.

Think:

Question	Chart
Count categories	Countplot
Trend over time	Line Chart
Relationship between numbers	Scatter Plot
Missing values	Heatmap
Distribution	Histogram
Comparison	Bar Chart

Chart 1

```
sns.countplot()
```

Question:

Which rating appears most?

Output:

```
TV-MA 10
TV-14  8
PG      2
```

Chart 2

```
plot(kind='line')
```

Question:

How many titles were added each year?

Output:

```
2016  —
2017  ———
2018  —————
2019  —————
```

Trend analysis.

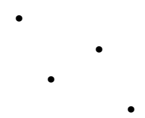
Chart 3

```
sns.scatterplot()
```

Question:

Are newer movies longer?

Each dot = one movie.



X-axis:

Release Year

Y-axis:

Runtime

Chart 4

```
sns.heatmap(df.isnull())
```

Question:

Where are missing values?

Bright area = many missing values.

Chart 5

```
value_counts().head(10)
```

Question:

Which countries create the most Netflix content?

USA

India
UK
Canada

Chart 6 (Important)

```
df.groupby(  
    ['year_added', 'type']  
).size().unstack()
```

This is the line you asked about before.

Step 1

Group by:

```
year_added  
type
```

Example:

year	type
2020	Movie
2020	Movie
2020	TV Show
2021	Movie

Step 2

```
.size()
```

Count rows.

Result:

year	type	count
2020	Movie	2
2020	TV Show	1
2021	Movie	1

Step 3

```
.unstack()
```

Convert:

year	type	count
2020	Movie	2
2020	TV Show	1

into:

year	Movie	TV Show
2020	2	1

Now easy to plot.

Chart 7

```
movies = df[df['type']=='Movie']
```

Means:

"Keep only movies."

Example:

Before:

```
Movie
Movie
TV Show
Movie
```

After:

```
Movie
Movie
Movie
```

Then:

```
sns.histplot(
movies['duration_int'],
bins=30,
```

```
kde=True
)
```

Question:

How are movie runtimes distributed?

Output:

```
90 min ██████████
100 min ██████████
120 min ██████████
150 min ██████████
```

KDE

The smooth curve drawn over bars.

Shows the overall shape of the distribution.

The One Thing To Memorize

Every data analysis project follows:

1. Import Libraries
2. Load Dataset
3. Explore Dataset
 - head()
 - shape
 - dtypes
 - isnull()
4. Clean Dataset
 - fillna()
 - dropna()
 - replace()
 - to_datetime()
5. Feature Engineering
 - create new columns
6. Analyze
 - groupby()
 - value_counts()
 - describe()
7. Visualize

- `countplot()`
- `lineplot()`
- `scatterplot()`
- `histplot()`
- `heatmap()`

8. Draw Conclusions

If you build 5–10 projects using this same flow, you'll start recognizing patterns and eventually be able to write most of the code yourself rather than memorizing individual lines.